Xuankun Yang

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Education

Shanghai Jiao Tong University — Shanghai, China
BEng in Artificial Intelligence — *Sep 2023 – Present*

- CGPA: 4.00 / 4.3, Rank 5 / 100
- Average score: 91.84 / 100, Rank 7 / 100

Research Interests

- Foundation models
- Mechanistic interpretability
- Efficient model training and inference

Experience

SJTU Artificial Intelligence Institute DeepVision Lab — Shanghai, China
Research Intern, PEFT — *Oct 2024 – May 2025*

- Mentor: Wei Shen
- Co-authored *Weight Spectra Induced Efficient Model Adaptation* (under review), a PEFT method motivated by empirical analysis of fine-tuning dynamics that modulates top singular directions for more effective adaptation.
- Validated the method across language, commonsense reasoning, and vision benchmarks including GLUE, Commonsense170K, and VTAB-1K, achieving strong performance under strict parameter budgets.

SJTU Artificial Intelligence Institute DeepVision Lab — Shanghai, China
Research Intern, Model Merging — *Jul 2025 – Present*

- Mentor: Wei Shen
- Conducted systematic research on model merging through extensive literature review and empirical investigation of existing paradigms.
- Wrote a technical survey categorizing model merging methods by optimization objective, parameter-space assumption, and interference handling strategy.

SJTU Artificial Intelligence Institute DeepVision Lab — Shanghai, China
Research Intern, Visual Token Pruning — *Oct 2025 – Present*

- Mentor: Wei Shen
- Co-first authored *Entropy-Aware Dense Visual Token Pruning* (under review), proposing dense semantic relevance modeling, entropy-assisted denoising, and submodular token selection with spatial priors to preserve fine-grained cues under strict token budgets.
- Demonstrated state-of-the-art accuracy-efficiency trade-offs across LLaVA, LLaVA-NeXT, LLaVA-Video, and Qwen2.5-VL families, with strong latency and FLOPs reduction under aggressive pruning ratios.

Papers

Weight Spectra Induced Efficient Model Adaptation
Second Author, Under review. — *May 2025*

Combating Textual Noise and Redundancy: Entropy-Aware Dense Visual Token Pruning
Co-first Author, Under review. — *Mar 2026*

Projects

A RAG Teaching Assistant System *Jan 2026*

- Built a multimodal RAG TA with multimodal embeddings, cross-modal retrieval, source-grounded citation, and a visible thinking mode to answer questions over course materials containing text, images, charts, and formulas.
- Constructed a small evaluation set and quantified both retrieval and generation quality.

Modular Reinforcement Learning Framework *Jan 2026*

- Developed a modular reinforcement learning framework for both discrete and continuous control, supporting DQNs, PPO and their variants across Atari and MuJoCo environments.
- Studied optimization stability, reward sensitivity, clipping ablations, and convergence behavior, building background for future work on RL-based LLM agent training and reasoning.

Data-Efficient Vision Transformer on CIFAR-10 *Jun 2025*

- Studied Vision Transformer on the small-sample CIFAR-10 setting, focusing on data-efficient learning; implemented native ViT and ResNet+ViT hybrid variants with configurable patch/head settings and stochastic depth.
- Reproduced and extended ViT-paper-style analyses on attention maps, attention distance, gradient rollout, embedding similarity, etc., and documented the findings in a technical blog.

End-to-End Voice Activity Detection Pipeline *May 2025*

- Built an end-to-end voice activity detection pipeline covering pre-emphasis, framing, windowing, acoustic feature extraction, frame-level classification, and time-domain speech segment restoration.
- Conducted extensive explorations over frame length/shift, preprocessing, smoothing, and model choices, and summarized the analysis in a technical blog.

Skills

Programming: Python, C++

Deep Learning: PyTorch, Hugging Face

Tools: Linux, Git, LaTeX, TensorBoard

LLM/VLM Frameworks: LLaVA, Qwen-VL